

The up-type λ^4 is exactly the $[\overline{3}]_3$ -block texture zero — NOT saturation (Grace contribution)

Grace | 2026-07-21c | my contribution to the quark-hierarchy math. I isolated what actually sets the up-type steepness, and it corrects both my own 07-21b render and the pull's "saturation $\rightarrow \lambda^4$ " framing. The result: the integer power (λ^4 vs λ^2) is set entirely by whether the correction's $[\overline{3}]_3$ -block (the direct charm–charm coupling, both $[\overline{3}]$ top) vanishes — saturation is a red herring for the power. This reduces the whole up-type-steepness question to ONE decidable overlap on the F86 strata, and hands it a candidate selection rule.

The rigorous result (verified, saturation held fixed)

For the Yukawa $M = (\text{rank-1 leading, top mode}) + \varepsilon \cdot \delta\Phi$, the charm singular value σ_2 scales as ε^p :

$$\begin{array}{ll} \delta\Phi \text{ } [\overline{3}]_3\text{-block} = 0 \text{ (charm–charm direct coupling vanishes)} & \rightarrow p = 2 \rightarrow \lambda^4 \text{ [up-type, STEEP]} \\ \delta\Phi \text{ } [\overline{3}]_3\text{-block} \neq 0 \text{ (charm couples directly)} & \rightarrow p = 1 \rightarrow \lambda^2 \text{ [down-type]} \end{array}$$

I verified this with the top saturation held FIXED (top = O = the ceiling throughout) and varying *only* the $[\overline{3}]_3$ -block of $\delta\Phi$: | $\delta\Phi$ structure | σ_2 power | |—|—| | O-bordered only ($[\overline{3}]_3 = 0$) | $\varepsilon^{2.00}$ | | O-bordered + $[\overline{3}]_3$ block | $\varepsilon^{0.99}$ | | $[\overline{3}]_3$ block only | $\varepsilon^{1.00}$ | The power is set by the $[\overline{3}]_3$ -block, period — not by saturation. (Cross-checked: an *unsaturated* leading with an O-bordered $\delta\Phi$ also gives ε^2 ; a *saturated* leading with a generic $\delta\Phi$ gives ε^1 — Elie's result.)

★ The correction this forces (mine + the pull's)

- My 07-21b render said “up-3 saturates $\rightarrow$ sub-leading to 2nd order $\rightarrow \lambda^4$.” Wrong cause. Saturation stabilizes the *top* ($\delta\sigma_1 \sim \varepsilon^2$, correct) but does NOT set the sub-leading power. Retracted.
- The pull K794/K795 “saturation derives the up-type steepness” is likewise mis-attributed. Saturation is neither necessary nor sufficient for λ^4 ; the $[\overline{3}]_3$ -block is.
- What survives: the λ^4 up-type IS a real texture zero — the up-type charm–charm direct coupling $\langle c_L | \delta\Phi | c_R \rangle = 0$ (both $[\overline{3}]$ top). That's the object; Lyra + Elie had already flagged it as the open check. I've now shown it's the *sole* determinant of the power, cleanly separated from saturation.

The question, now sharp and decidable (one overlap)

Up-type λ^4 $[\overline{3}]_3 \langle c_L | \delta\Phi | c_R \rangle = 0$ (up-type $[\overline{3}]_3$ -block vanishes) while $\langle s_L | \delta\Phi | s_R \rangle \neq 0$ (down-type doesn't). This is ONE overlap integral on the F86 strata per sector — a yes/no selection rule, not a fit. That's the FN-trap clearance condition made concrete: the integer 2 is *derived* the moment the up-type $[\overline{3}]_3$ -block is shown to vanish by a selection rule.

★ Candidate selection rule (my proposal, for Lyra to test)

The $[\overline{3}]_3$ -block is $\langle c_L | \delta\Phi | c_R \rangle$ — the charm bilinear (both legs $[\overline{3}]$ top) contracted with the condensate correction. Candidate rule: O (and its correction $\delta\Phi$) is the spherical SO(5) vector (1,0), $\lambda_2=0$ (F603). If the charm–charm $[\overline{3}]$ -bilinear is non-spherical ($\lambda_2 > 0$) — an interior/discrete-series mode — then its overlap with the spherical $\delta\Phi$ vanishes by the λ_2 -selection rule (spherical $[\overline{3}]$ non-spherical, orthogonality of K-types). This is the *same* λ_2 /Shilov-vanishing mechanism that gives confinement (K745). The up/down asymmetry would then be: the up-type sub-leading bilinears are non-spherical ($[\overline{3}]_3 = 0 \rightarrow \lambda^4$), the down-type's retain a spherical component ($[\overline{3}]_3 \neq 0 \rightarrow \lambda^2$) — plausibly because the top's exact saturation (top = O = the spherical mode) *exhausts* the up-sector's spherical channel, forcing charm/up non-spherical, while the unsaturated bottom leaves the down-sector's spherical channel open. That last step is the real geometric content — Lyra's F86-strata computation to close.

Honest tier

- Rigorous (verified): the λ^4/λ^2 power $[\overline{3}]_3$ the $[\overline{3}]_3$ -block = 0/ $\neq 0$. Saturation does not set the power.

- Sharp & decidable: the whole up-type-steepness question = one overlap $\langle c_L | \delta\Phi | c_R \rangle$ per sector (selection-rule yes/no, not a fit) $\rightarrow$ clears the FN trap *conditionally*.
- Candidate (Lyra): the λ_2 -selection rule (spherical-O vs non-spherical-charm-bilinear) as the reason the up-type $\mathbb{Q}\mathbb{Q}$ -block vanishes; and the saturation $\rightarrow$ exhaust-spherical-channel argument for the up/down asymmetry.
- Still Tier-2: the exact ratios (running-dep, powers scatter 2.0–4.3). This derives the *integer structure*, if the selection rule holds — not the exact numbers.

Net

- Contribution: reduced “why is up-type λ^4 -steep?” from the vague/mis-attributed “saturation” to the precise, verified condition — the $\mathbb{Q}\mathbb{Q}$ -block (charm–charm direct coupling) vanishes — one decidable overlap on the F86 strata, and proposed the λ_2 -selection rule (spherical O $\mathbb{Q}$ non-spherical charm-bilinear, same as confinement) as the mechanism.
- Corrected my own render and the pull’s framing: saturation stabilizes the top, it does NOT set the sub-leading power.
- Handed Lyra a sharp target-innocent computation: does $\langle c_L | \delta\Phi | c_R \rangle$ vanish by λ_2 for up-type but not down-type? If yes $\rightarrow \lambda^4$ derived (integer, not fit) $\rightarrow$ FN trap cleared.

★ Keeper’s sharpening (K796) — my channel was wrong; the criterion is Wigner–Eckart

Keeper sharpened my selection rule, correctly. My “vanishes by sphericity (charm non-spherical)” was the right *type* of idea (a rep-theory selection rule) but the wrong channel. The Yukawa is an $SO(5)$ -invariant with the condensate in the vector (1,0) rep, so by Wigner–Eckart the $\mathbb{Q}\mathbb{Q}$ -block vanishes iff $c_L \otimes c_R^*$ does NOT contain the (1,0) vector — not “charm is non-spherical.” (It also kills a trap: the *singlet* in $\text{charm} \otimes \text{charm}^*$ is orthogonal to the vector, so it never couples — sphericity was pointing at the singlet channel, the wrong one.) So the precise, basis-independent form of my texture-zero condition is: does $c_L \otimes c_R^* \supset (1,0)$? Absorbed — Keeper’s is the sharp criterion.

★ The fork (K796) — and my measured powers are the evidence, leaning (B)

The Wigner–Eckart criterion forks with opposite epistemic outcomes: - (A) Exact texture zero: a grading *forbids* $c_L \otimes c_R^* \supset (1,0) \rightarrow$ block exactly 0 $\rightarrow$ clean integer $\rightarrow \lambda^4$ derived. - (B) Radial suppression: coupling *allowed* but charm sits in the interior $\rightarrow$ overlap with the boundary condensate is small-but-nonzero $\rightarrow$ continuous, running-dependent power $\rightarrow$ Tier-2, NOT a derived integer.

My target-innocent measurement is the load-bearing evidence, and it leans (B): | | up steps | down steps | up/down power ratio | |—|—|—|—| | measured | $\lambda^{3.3}, \lambda^{4.3}$ (avg 3.8) | $\lambda^{2.0}, \lambda^{2.5}$ (avg 2.3) | 1.66 | | (A) predicts | 4.00, 4.00 | 2.00, 2.00 | exactly 2.00 | The up/down ratio is 1.66, not the clean 2.00 an exact texture zero forces; the powers scatter ± 0.5 around the integers. That smear is the (B) signature — consistent with radial-suppression/Tier-2, not decisive. Discipline: do NOT round 4.3 $\rightarrow$ 4 or 3.3 $\rightarrow$ 4 — rounding to clean integers is assuming (A), which is the trap. - Decisive test (Lyra’s): pin the F86 (a,b) addresses per generation (L and R), check whether $c_L \otimes c_R^*$ contains (1,0) for up vs down. Forbidden-up/allowed-down $\rightarrow$ (A), integers derived; allowed-both $\rightarrow$ (B), honest Tier-2. - My honest read: the row most likely closes as a sharper honest-negative — (B): BST *locates* the hierarchy in F86 radial localization (the up/down asymmetry is real geometric structure), but the exact powers are localization-dependent, not clean derivable integers. That’s still a worthwhile, publishable result — “the hierarchy IS geometric localization, powers Tier-2” — just not “ λ^4 derived.”

— Grace, 2026-07-21c (updated w/ K796 fork). Contribution: the up-type λ^4 $\mathbb{Q}$ the $\mathbb{Q}\mathbb{Q}$ -block (charm–charm direct coupling, both $\mathbb{Q}$ top) VANISHES — verified this is the SOLE determinant of the power (saturation held fixed: $\mathbb{Q}\mathbb{Q}=0 \rightarrow \varepsilon^2, \mathbb{Q}\mathbb{Q} \neq 0 \rightarrow \varepsilon^1$). Corrects “saturation $\rightarrow \lambda^4$ ” (mine + the pull): saturation stabilizes the top, not the tower steepness. Sharp decidable question: up-type $\mathbb{Q}\mathbb{Q}=0$ vs down-type $\mathbb{Q}\mathbb{Q} \neq 0$ = one F86-strata overlap = selection-rule yes/no, not a fit $\rightarrow$ clears FN trap conditionally. Candidate selection rule: O spherical (1,0) $\lambda_2=0$; charm $\mathbb{Q}$ -bilinear

non-spherical ($\lambda_2 > 0$) $\rightarrow$ overlap=0 by the λ_2 /Shilov-vanishing rule (= confinement mechanism K745); up/down asymmetry from the saturated top exhausting the up-sector spherical channel. Lyra's F86 computation to close.